

# Walmart Sales Data Analysis using SQL

## Business Problems and Purpose

**Q1. What are the different payment methods, and how many transactions and items were sold with each method?**

**Purpose:** Understand customer payment preferences by comparing transaction counts and total items sold for each payment method.

**Q2. Which category received the highest average rating in each branch?**

**Purpose:** Identify the highest-rated product category in every branch based on customer ratings.

**Q3. What is the busiest day of the week for each branch based on transaction volume?**

**Purpose:** Determine the busiest shopping day for each branch to improve staffing and inventory planning.

**Q4. Calculate the total quantity sold by each payment method.**

**Purpose:** Analyze which payment method is associated with the highest quantity of products sold.

**Q5. Determine the average, minimum, and maximum rating of categories for each city.**

**Purpose:** Evaluate customer satisfaction for every product category across different cities.

**Q6. Calculate the total profit by category and rank the categories from highest to lowest profit.**

**Purpose:** Identify the most profitable categories and compare their contribution to overall profit.

**Q7. Determine the most common payment method for each branch.**

**Purpose:** Find the preferred payment method used by customers in every branch.

**Q8. Analyze sales shifts throughout the day by counting the number of transactions in each shift for every branch.**

**Purpose:** Identify the busiest sales shift (Morning, Afternoon, Evening) for each branch.

**Q9. Identify the five branches with the highest decrease ratio in revenue compared to the previous year (2025 vs 2026).**

**Purpose:** Highlight branches with the largest year-over-year revenue decline.

**Q10. Find the percentage of total transactions contributed by each payment method.**

**Purpose:** Measure the contribution of each payment method to the total number of transactions.

**Q11. Find the best-selling category in every branch based on the total quantity sold.**

**Purpose:** Identify the most popular product category in each branch.

**Q12. Rank all branches based on their total revenue from highest to lowest.**

**Purpose:** Compare branch performance using total revenue.

**Q13. Which month generated the highest total revenue?**

**Purpose:** Determine the best-performing month based on revenue.

**Q14. Find the branches whose revenue increased by more than 20% from 2025 to 2026.**

**Purpose:** Identify branches with strong year-over-year revenue growth.

**Q15. Create an Executive Dashboard that provides a one-page summary of Walmart's business performance.**

**Purpose:** Summarize key business KPIs such as total revenue, profit, transactions, ratings, top branch, top category, top city, and most-used payment method.